VISHNU
UNIVERSAL LEARNING

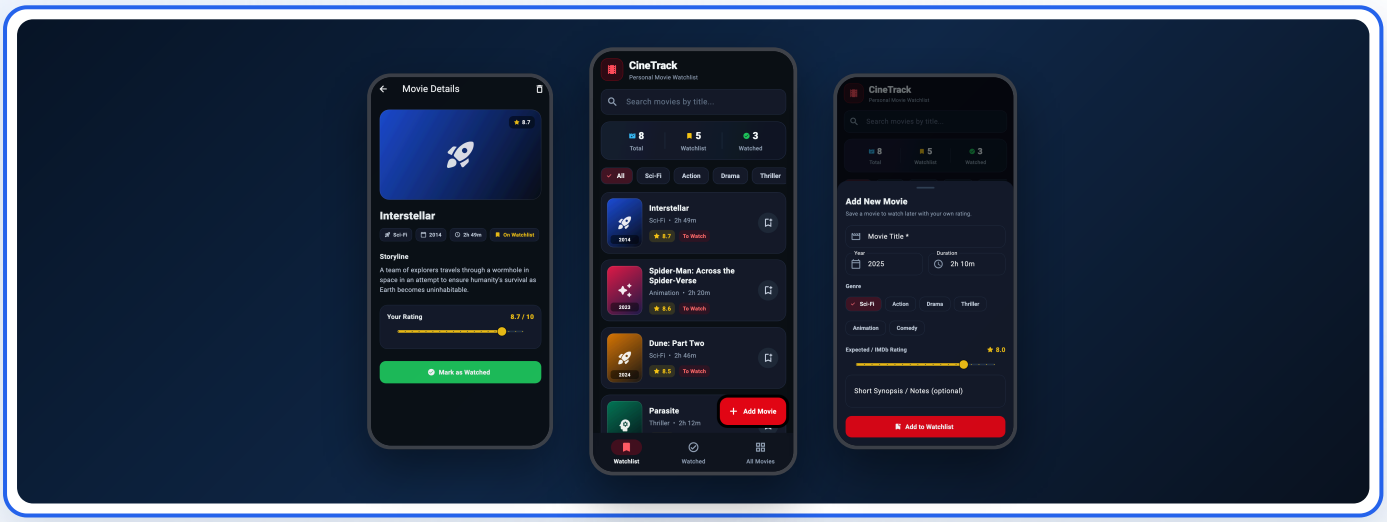
VISHNU INSTITUTE OF TECHNOLOGY

Bhimavaram, Andhra Pradesh, India

FLUTTER PROJECT REPORT

CineTrack – Movie Watchlist App

A Modern Dark-Themed Movie Watchlist Mini Project Built Using Flutter



Submitted By:

Name	: **KHANDAVALLI SIMON JOSEPH**
Register Number	: 24PA1A05B6
Branch	: CSE
Year	: 3rd Year
Section	: B
Academic Year	: 2026–2027

Project Guide:

Not Provided

Flutter
Build • Design • Innovate

Submitted in partial fulfillment of the requirements for the award of Bachelor of Technology in Computer Science and Engineering

Academic Year: 2026 – 2027

1. Project Overview

This report presents **CineTrack**, a clean, dark-themed Movie Watchlist mini-project developed using Flutter and Dart. The goal of the application is to help users organize movies they plan to watch and movies they have already watched inside an intuitive, distraction-free interface. Key features include live title search, genre-based filtering, summary statistics, a movie detail view with interactive rating adjustment, and a quick add-movie modal sheet.

Student Details

Student Name	KHANDAVALLI SIMON JOSEPH
Register Number	24PA1A05B6
Branch / Year / Section	CSE / 3rd Year / B
Academic Year	2026–2027
Project Guide	Not Provided
Technology	Flutter and Dart (Material 3 Dark Theme)

Project Objectives

- Build a clean, visually appealing dark-theme movie watchlist interface using Flutter Material 3 widgets.
- Separate movies cleanly into *Watchlist*, *Watched*, and *All Movies* views via bottom navigation.
- Provide instant title search and horizontal genre filtering chips for quick movie discovery.
- Allow users to add new movies, view storylines, update personal ratings, and toggle watch status.
- Keep the codebase modular, readable, and easy to explain using standard `StatefulWidget` state management.

Scope and Implementation Status

This mini-project has been implemented, analyzed with `flutter analyze` (zero issues), and verified with automated widget tests (`flutter test`). All screenshots in this report are captured directly from the rendered Flutter application widgets. Movie entries are preloaded in local memory for offline simplicity without requiring an external backend server or API key.

2. Architecture and Data Flow

The CineTrack application follows a clean, beginner-friendly presentation-oriented structure. Flutter widgets render the dark cinema UI, standard `StatefulWidget` and `setState()` manage interactive list updates, and a simple local data module supplies the initial movies.

Application Architecture

USER	Movie Enthusiast / Student
PRESENTATION LAYER	HomeScreen • MovieDetailScreen • MovieCard • StatsBanner • AddMovieSheet
NAVIGATION & STATE	Material 3 NavigationBar • Navigator.push (Hero) • StatefulWidget (setState)
DATA & MODEL LAYER	Movie Model (`movie.dart`) • Preloaded Sample Movies (`sample_movies.dart`)

Basic User Flow

1. User opens the CineTrack home screen and views the live stats banner and Watchlist movies.
2. User searches by movie title or taps genre chips (*Sci-Fi, Action, Drama, Thriller, Animation, Comedy*).
3. User taps the bookmark button on any card to move a movie between Watchlist and Watched.
4. User taps a movie card to open `MovieDetailScreen` with a Hero poster animation and rating slider.
5. User taps the **+ Add Movie** button to open the bottom sheet and save a new movie.

Design Choice

Flutter was chosen for its fast UI iteration, built-in Material 3 dark theming, and smooth `Hero` screen transitions. Using plain `StatefulWidget` with `setState()` keeps the architecture transparent and straightforward to understand, maintain, and demonstrate during academic evaluations.

3. Project Structure and Modules

The Flutter project is organized into focused folders separating the data model, sample dataset, screens, and reusable UI components inside `cinetrack_watchlist/`.

```
cinetrack_watchlist/
  lib/
    main.dart
    models/
      movie.dart
    data/
      sample_movies.dart
    screens/
      home_screen.dart
      movie_detail_screen.dart
    widgets/
      stats_banner.dart
      movie_card.dart
      add_movie_sheet.dart
  test/
    widget_test.dart
    screenshot_test.dart
  screenshots/
    home_watchlist.png
    movie_details.png
    add_movie_sheet.png
  pubspec.yaml
```

Important Modules

Module	Purpose
main.dart	App entry point (`CineTrackApp`) configuring the dark Material 3 color palette and widget themes.
movie.dart	Defines the `Movie` class, `copyWith()` helper, genre icons, and poster gradient colors.
sample_movies.dart	Provides the genre list and 8 preloaded sample movies across Watchlist and Watched tabs.
home_screen.dart	Manages movie state, search bar, genre filters, bottom tabs, and floating action button.
movie_detail_screen.dart	Displays full movie details, storyline, interactive 1–10 rating slider, watch toggle, and delete dialog.
movie_card.dart	Reusable dark card widget with gradient poster thumbnail, gold star badge, status pill, and quick toggle.
add_movie_sheet.dart	Modal bottom sheet form with validation, genre chips, and rating slider to add new movies.
stats_banner.dart	Compact 3-column summary banner showing Total, Watchlist, and Watched movie counts.

4. Representative Flutter Code

The excerpt below from `lib/main.dart` and `lib/models/movie.dart` illustrates how the CineTrack application initializes the Material 3 dark cinema theme and models each movie entry cleanly.

```
import 'package:flutter/material.dart';
import 'screens/home_screen.dart';

void main() {
  runApp(const CineTrackApp());
}

/// Root widget of the CineTrack Movie Watchlist application.
class CineTrackApp extends StatelessWidget {
  const CineTrackApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'CineTrack – Movie Watchlist',
      debugShowCheckedModeBanner: false,
      themeMode: ThemeMode.dark,
      darkTheme: ThemeData(
        useMaterial3: true,
        brightness: Brightness.dark,
        scaffoldBackgroundColor: const Color(0xFF0B0F17),
        colorScheme: const ColorScheme.dark(
          primary: Color(0xFFE50914),
          secondary: Color(0xFFF5C518),
          surface: Color(0xFF141B29),
        ),
      ),
      home: const HomeScreen(),
    );
  }
}
```

Code Explanation

- `main()` is the Dart entry point that launches the Flutter widget tree via `runApp()`.
- `CineTrackApp` configures `MaterialApp` with `ThemeMode.dark` and Material 3 enabled.
- `scaffoldBackgroundColor` (`#0B0F17`) and surface (`#141B29`) establish the dark cinema look, paired with crimson primary and IMDb-gold secondary accents.
- `HomeScreen` is set as the initial route, where `setState()` updates search, filters, and movie lists dynamically.
- `Movie` uses immutable fields and a `copyWith()` helper so status and rating updates remain clean and predictable.

5. Features and User Flow

User Interaction Flow

- Launch the app to immediately view saved Watchlist movies alongside Total, Watchlist, and Watched counters.
- Type any keyword into the search bar or tap a genre chip to narrow down the movie list in real time.
- Tap the right-side bookmark/check icon on any `MovieCard` to toggle between *To Watch* and *Watched*.
- Tap a movie card to navigate to `MovieDetailScreen`, adjust its 1–10 star rating, or remove it from the list.
- Tap **+ Add Movie** to open the modal bottom sheet, enter movie details, pick a genre, and save it.

Implemented Interface Features

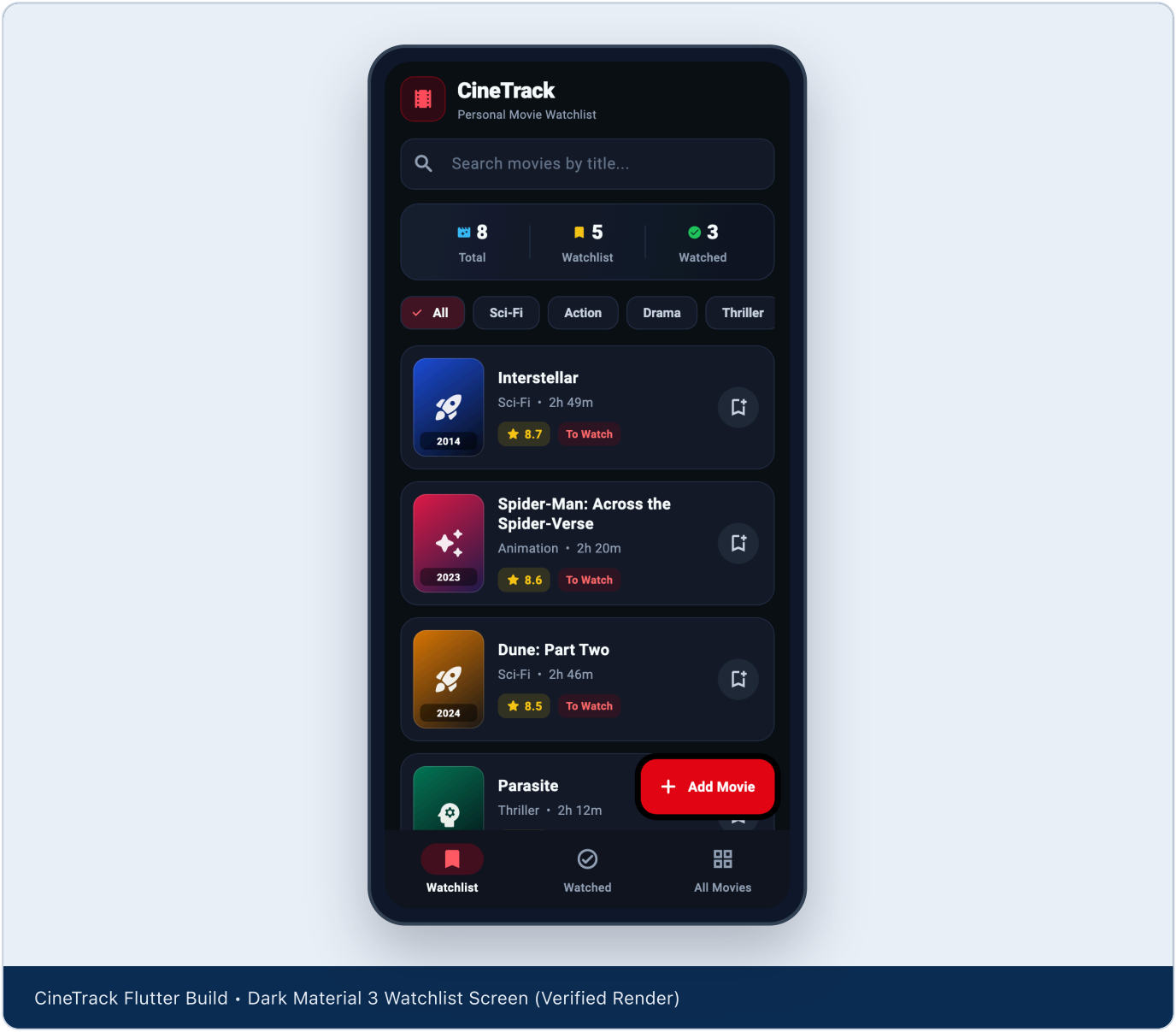
- **Dark Cinema Palette:** Deep obsidian background, subtle slate borders, crimson CTAs, emerald watched badges, and gold rating pills.
- **Stylized Gradient Poster Tiles:** Genre-matched gradients with Material icons and release year badges that work 100% offline.
- **Smooth Hero Transitions:** Poster thumbnails animate seamlessly from the list card into the detail banner.
- **Form Validation & Feedback:** Required title check on the Add Movie form plus floating `SnackBar` confirmations.
- **Empty State Handling:** Friendly fallback message when a tab or search filter matches no movies.

Current Scope & Future Enhancements

In its current mini-project version, CineTrack stores movies in memory starting from 8 curated sample titles, keeping the project lightweight and dependency-free. As future enhancements, the app can integrate local storage using `shared_preferences` or SQLite, fetch live movie posters via the TMDB API, and support custom watchlists.

6. Application Output – Watchlist Home Screen

Figure 1 shows the actual rendered Flutter dark-theme home screen of CineTrack. It combines the branded header, live search field, 3-metric summary banner (Total, Watchlist, Watched), scrollable genre chips, and gradient-poster movie cards above the 3-tab bottom navigation bar.



CineTrack Flutter Build • Dark Material 3 Watchlist Screen (Verified Render)

Figure 1. CineTrack dark-theme Watchlist home screen with live stats, search bar, genre filters, and movie cards.

7. Application Output – Details & Add Movie

Figure 2 illustrates the secondary interactive workflows of CineTrack: the `MovieDetailScreen` with its Hero poster banner, metadata badges, storyline, and interactive rating slider (left), alongside the `AddMovieSheet` modal bottom sheet for saving new movies (right).

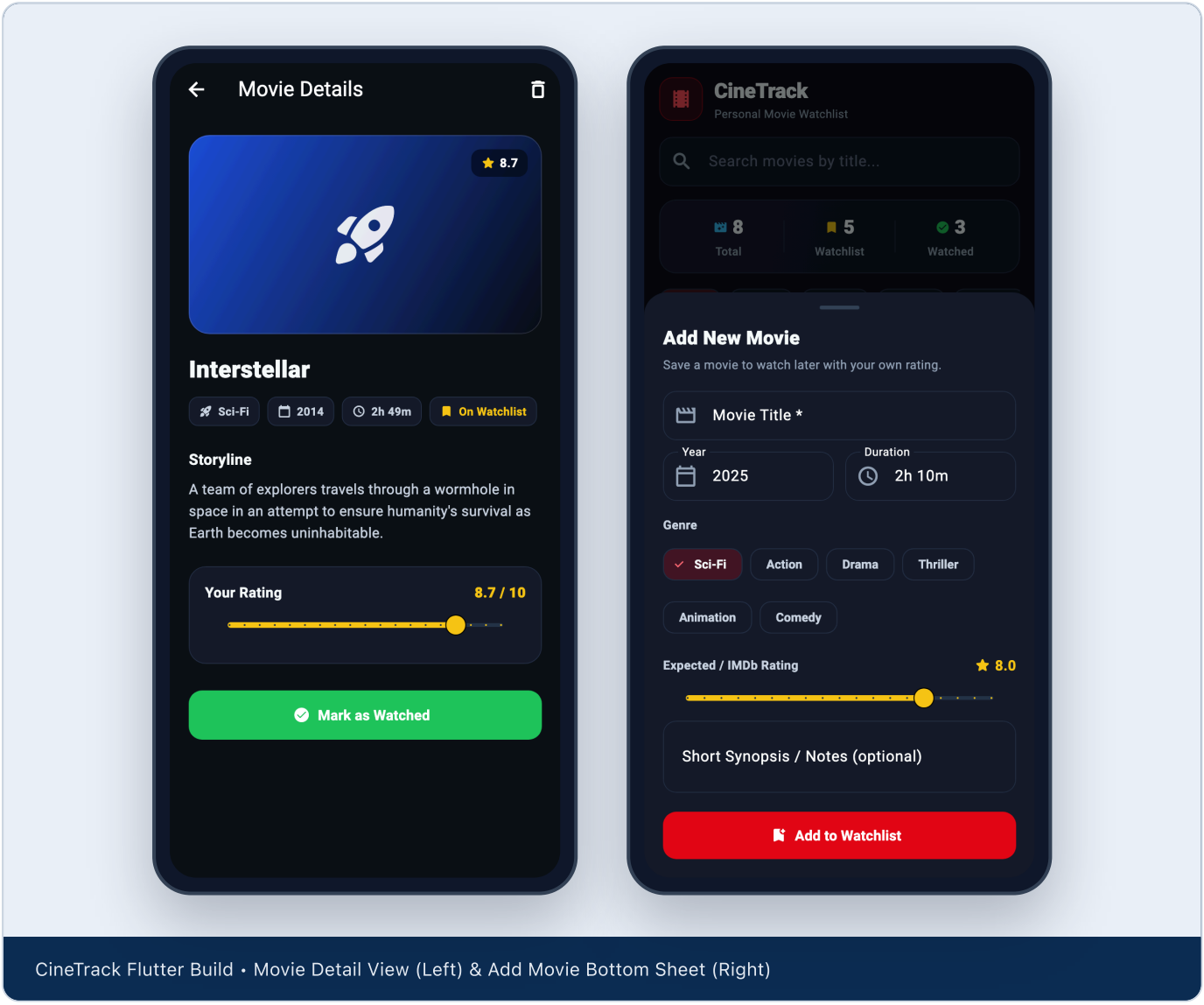


Figure 2. Movie Details screen with interactive rating control (left) and Add New Movie modal sheet (right).

8. Conclusion and Development Setup

The **CineTrack – Movie Watchlist App** demonstrates how a clean, modern mobile interface can be built in Flutter without complex dependencies. By combining a cohesive dark Material 3 palette, modular widgets, and straightforward `StatefulWidget` state management, the project delivers a polished user experience while remaining easy to read, explain, and extend.

Tools and Technologies Used

- **Flutter SDK (3.47.2)** – cross-platform UI toolkit and rendering engine.
- **Dart (3.13.2)** – strongly typed application programming language.
- **Material 3 Dark Components** – `NavigationBar`, `ChoiceChip`, `Slider`, `ModalBottomSheet`, and `Hero`.
- **VS Code / Android Studio** – development, formatting, and static analysis environment.
- **flutter_test** – automated widget testing and UI verification.

Build and Run Procedure

1. Ensure Flutter SDK is installed and available on your system `PATH`.
2. Clone the repository:
   ```
   git clone https://github.com/SimonJoe062209/cinetrack-flutter-watchlist.git
   ```
3. Navigate into the project directory and fetch dependencies:
   ```
   cd cinetrack-flutter-watchlist && flutter pub get
   ```
4. Run static analysis and automated widget tests:
   ```
   flutter analyze && flutter test
   ```
5. Launch the application on an emulator, Chrome, or connected device:
   ```
   flutter run
   ```

Verification Summary: Static analysis completed with zero warnings or errors, and all widget tests covering watchlist rendering, tab switching, search filtering, and adding movies passed successfully.

9. GitHub Repository

Project Title

CineTrack – Movie Watchlist App (Flutter Mini Project)

Repository Link

https://github.com/SimonJoe062209/cinetrack-flutter-watchlist

Repository Contents

The GitHub repository hosts the complete source code for the CineTrack Flutter project, including the `lib/` application modules (`models` , `data` , `screens` , `widgets`), automated widget tests in `test/` , verified UI screenshots in `screenshots/` , and setup instructions in `README.md` .

End of Report

Student: KHANDAVALLI SIMON JOSEPH | Register Number: 24PA1A05B6
Department of Computer Science and Engineering
Vishnu Institute of Technology • Academic Year 2026–2027